

|    |   |                                                                                                                                                                                                                                                                                                                 |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | Also, centripetal force ( $r = mv^2/r$ ) is greater for P than Q as $v_1 > v_2$ .                                                                                                                                                                                                                               |
| 10 | B | <p>At P: <math>N + mg = mv^2/r</math><br/> <math>N = mv^2/r - mg</math><br/> <math>v = \sqrt{rg}</math> when <math>N \approx 0</math></p> <p>Since GPE at P and Q is the same, KE at P = KE at Q.</p> <p>At Q: <math>mg - N = mv^2/2r</math><br/> <math>N = mg - m(rg)/2r</math><br/> <math>N = mg/2</math></p> |
| 11 | D | By conservation of energy                                                                                                                                                                                                                                                                                       |